Nationwide TDP Capstone Project 2026 V2.0

Sprint 3 — Customer Insights & Recommendations

Duration: 3 Days

Objective

Extend the carbon-footprint MVP to provide customers with meaningful **carbon insights, spending trends and practical recommendations**.

Sprint Goal

A customer can view their carbon impact, understand spending patterns and receive practical recommendations that could reduce both **carbon impact and costs**.

1. Agile & Team Working

- Review and revise the team's **Definition of Ready (DoR)** and **Definition of Done (DoD)**.
- Update the product and sprint backlogs.
- Maintain the task board and development log.
- Review team roles, responsibilities and collaboration.

2. Customer Carbon Insights

Provide clear information such as:

- Total estimated carbon impact.
- Carbon impact by spending category.
- Highest-impact spending categories.
- Changes in carbon impact over time.
- Spending patterns contributing to carbon impact.

3. Recommendations

Develop practical recommendations based on customer spending behaviour.

Recommendations should identify:

- Potential lower-carbon alternatives.
- Spending behaviours that could be changed.
- Practical actions the customer can take.

- Potential cost-saving opportunities.
- The relationship between **financial savings and carbon reduction**.

Recommendations should be based on validated data and transparent business rules.

4. API Development

Extend the API to support customer insights and recommendations:

GET /health

GET /categories

POST /spend

POST /carbon-impact

Additional endpoints may be introduced where useful for:

- Carbon summaries.
- Trends.
- Recommendations.
- Potential savings.

5. Data & Analysis

Apply relevant learning from:

- **Fundamentals of Data Literacy**
- **Querying SQL Databases using T-SQL**
- **Introduction to Data Science for Data Professionals**
- **Data Handling in Python**
- **Databricks Data Engineering** — stretch

Ensure calculations and insights are based on validated data.

6. Testing

Extend testing to cover:

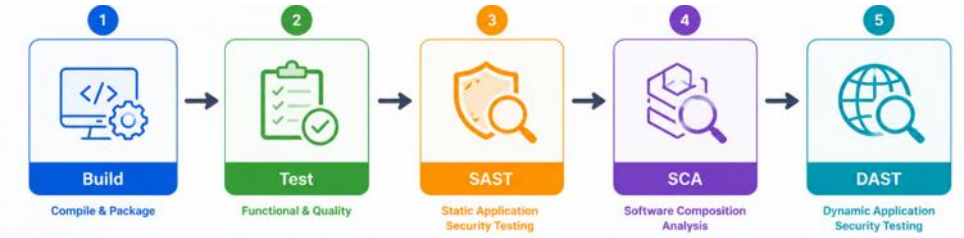
- Carbon summaries.
- Category calculations.
- Trends.
- Recommendation logic.

- Invalid or incomplete data.
- API responses.
- Database integration.

Include unit and integration tests as appropriate.

7. DevSecOps

Continue the pipeline:



Focus on:

- Secure customer-data handling.
- Input validation.
- Secure API design.
- Dependency security.
- Secrets management.
- Security testing.
- Remediation of vulnerabilities.

8. Responsible AI Assistance

AI can assist with:

- Code development.
- Test generation.
- Data analysis.
- SQL queries.
- Recommendation development.
- Troubleshooting.
- Documentation.

Teams must **review, understand, test and security-check** AI-generated output.

AI recommendations should be validated before being presented to customers.

9. Sprint 3 Showcase

Demonstrate:

1. Customer carbon summary.
2. Carbon impact by category.
3. Spending/carbon trends.
4. Highest-impact categories.
5. Practical recommendations.
6. Potential cost-saving opportunities.
7. Validated data and calculations.
8. Automated tests passing.
9. CI/CD pipeline operational.
10. SAST and SCA completed.
11. DAST completed where practical.

Sprint 3 Success

The customer can now **understand their carbon impact and spending patterns and receive practical recommendations that could reduce both carbon emissions and costs**, supported by tested application logic, validated data and secure DevSecOps practices.